

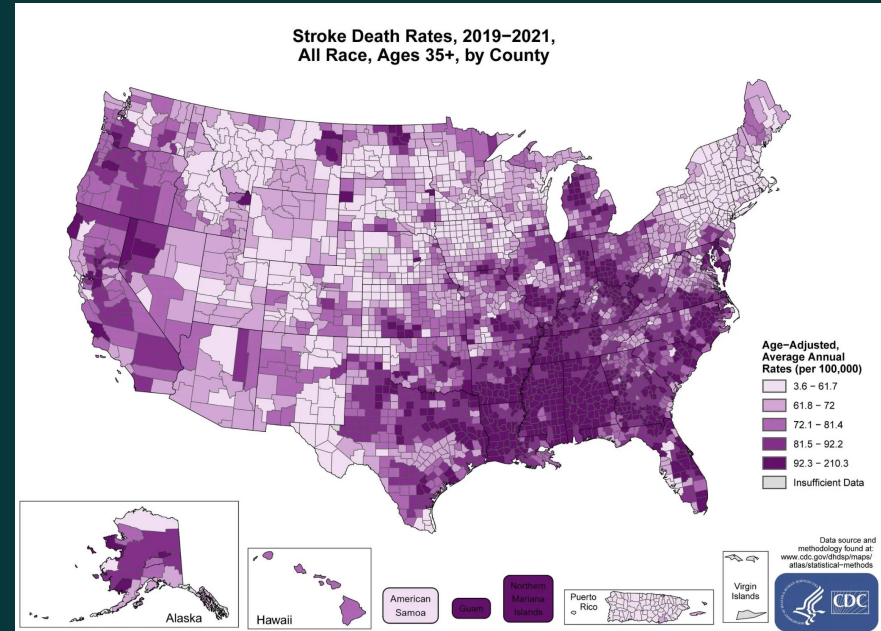
StrokeShield Analytics

Identifying Stroke Risk to Enable Preventive Care



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Why This Matters



Dataset Overview

	Column	Description	Feature Type	Valid Values/Range
0	id	Unique patient identifier assigned to each individual in the dataset	Identifier	67 to 72,940 (5,110 unique)
1	gender	Biological sex of the patient	Categorical - Nominal	Male (2,115), Female (2,994), Other (1)
2	age	Patient's age in years at time of data collection	Continuous	0.08 to 82.00 years (Mean: 43.23)
3	hypertension	Binary indicator of hypertension diagnosis (high blood pressure $\geq 140/90$ or $\geq 130/80$ mmHg)	Binary	0 = No (4,612), 1 = Yes (498)
4	heart_disease	Binary indicator of cardiovascular disease (CAD, heart failure, MI, AFib, valvular disease)	Binary	0 = No (4,834), 1 = Yes (276)
5	ever_married	Marital status indicator - whether the patient has ever been married	Categorical - Nominal (Binary)	Yes (3,353), No (1,757)
6	work_type	Patient's employment category or occupational status	Categorical - Nominal	Private (2,925), Self-employed (819), children (687), Govt_job (657), Never_worked (22)
7	Residence_type	Type of residential area where patient lives	Categorical - Nominal (Binary)	Urban (2,596), Rural (2,514)
8	avg_glucose_level	Average blood glucose level in mg/dL (likely fasting glucose or HbA1c equivalent)	Continuous	55.12 to 271.74 mg/dL (Mean: 106.15)
9	bmi	Body Mass Index calculated as $\text{weight(kg)} / \text{height(m)}^2$	Continuous	10.30 to 97.60 (Mean: 28.89)
10	smoking_status	Patient's smoking history and current status	Categorical - Nominal	never smoked (1,892), Unknown (1,544), formerly smoked (885), smokes (789)
11	stroke	Target variable indicating whether the patient experienced a stroke	Binary (TARGET)	0 = No stroke (4,861), 1 = Stroke (249)

Addressing Data Quality Concerns

- “Unknown” in Smoking Status column—missing data or valid category?
- BMI column contains missing values as well
- Checked for impossible values

Columns with missing or null values

id	0
gender	0
age	0
hypertension	0
heart_disease	0
ever_married	0
work_type	0
Residence_type	0
avg_glucose_level	0
bmi	201
smoking_status	1544
stroke	0
dtype:	int64

Percentage of each column that is missing data

id	0.000000
gender	0.000000
age	0.000000
hypertension	0.000000
heart_disease	0.000000
ever_married	0.000000
work_type	0.000000
Residence_type	0.000000
avg_glucose_level	0.000000
bmi	3.933464
smoking_status	30.215264
stroke	0.000000
dtype:	float64

Stroke Rate Analysis By Life Stage

- **Uneven bin sizes.**
- **Stroke rates tend to remain very low in early life stages**
- **The jump between Young Adult and Middle Adult groups (age 40) is the most dramatic.**
- **Good for communicating risk to a general audience.**

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STROKE RATE ANALYSIS BY AGE GROUP
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1. DETAILED STROKE STATISTICS BY AGE GROUP:
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	Total Patients	Stroke Cases	Stroke Rate (%)
age_group			
Child (0-12)	588	1	0.17
Adolescent (13-19)	378	1	0.26
Young Adult (20-39)	1204	6	0.50
Middle Adult (40-59)	1564	60	3.84
Older Adult (60-74)	858	79	9.21
Elderly (75+)	518	102	19.69

Stroke Rate Analysis

WHO Standard Age Categories

- Groups have clinical meaning
- Bin sizes better suited for rapidly changing children.
- Uneven bin-sizes
- Broader adult categories

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Stroke Cases in each group
age_group
Infant/Toddler (0-4)      1
Child (5-14)             1
Youth (15-24)            0
Adult (25-44)            8
Middle Age (45-64)      80
Young-Old (65-74)       57
Old-Old (75-84)        102
Oldest-Old (85+)        0
dtype: int64

age_group
Infant/Toddler (0-4)      255
Child (5-14)             444
Youth (15-24)            537
Adult (25-44)           1297
Middle Age (45-64)      1550
Young-Old (65-74)       509
Old-Old (75-84)         518
Oldest-Old (85+)         0
Name: count, dtype: int64

Percentage of Patients Who Had a Stroke by Age Group:
age_group
Infant/Toddler (0-4)      0.39
Child (5-14)             0.23
Youth (15-24)            0.00
Adult (25-44)            0.62
Middle Age (45-64)       5.16
Young-Old (65-74)       11.20
Old-Old (75-84)         19.69
Oldest-Old (85+)         NaN
Name: stroke, dtype: float64
```

Stroke Rate Analysis By Decade

Key findings:

- Even bin sizes
- Stroke rate exceeds 5% at 50-59
- Stroke rate exceeds 10% at 60-69

	Total Strokes	Stroke Rate (%)
age_group		
0-9	1	0.21
10-19	1	0.20
20-29	0	0.00
30-39	6	0.92
40-49	12	1.64
50-59	48	5.76
60-69	47	7.57
70-79	94	16.52
80+	40	21.51

```
Patients per Age Group:  
age_group  
0-9      472  
10-19    494  
20-29    549  
30-39    655  
40-49    730  
50-59    834  
60-69    621  
70-79    569  
80+      186  
Name: count, dtype: int64
```

Strongest Risk Factors

1. Age – unmodifiable
2. Heart disease – depends (unmodifiable/modifiable)
3. Hypertension – modifiable

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Stroke Rate (%) by Heart Disease:
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heart_disease
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1      17.028986
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0       4.178734
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```
Name: stroke, dtype: float64  
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```

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Stroke Rate (%) by Hypertension:
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hypertension
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1      13.253012
```

```
0       3.967910
```

```
Name: stroke, dtype: float64  
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```

```
Stroke Rate (%) by Age Group:
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```
age_group
```

```
0-9      0.21
```

```
10-19    0.20
```

```
20-29    0.00
```

```
30-39    0.92
```

```
40-49    1.64
```

```
50-59    5.76
```

```
60-69    7.57
```

```
70-79   16.52
```

```
80+     21.51
```

Recommendation:

**Prevention programs should target
individuals in their late forties to
early fifties**

THANK YOU!